

XScientist Board: Artifact-Routed Submission-Path Control for Autonomous Research Systems

Anonymous Authors

Abstract

Autonomous research pipelines can now generate plausible papers, but plausibility is not the same as submission readiness. The missing capability is often *submission hardening*: deciding which claims may advance, which defects must remain visible as debt, and which owner must repair each blocker before a paper is treated as reviewer-defensible. We study this late-stage systems problem and present **XScientist Board**, an artifact-routed submission-hardening architecture that treats paper finalization as a promotion-and-repair policy over explicit artifacts, ownership, and promotion rules rather than as a monolithic rewrite loop. The architecture couples (i) artifact-gated manuscript promotion via claim–evidence bindings, (ii) specialized writing lanes with bounded authority, and (iii) a read-only adversarial critic whose findings must route into owner-specific, verifiable repairs. The contribution is therefore a control policy over artifact state transitions, not a claim that more agents alone improve paper quality. We implement the design in the XScientist codebase and evaluate it on a frozen 10-topic artifact replay suite that holds topics and idea fixtures fixed across workflows. Compared to `classic_pipeline`, `multi_agent_board` improves mean ready rate from 0.00 to 0.80, reduces unsupported claims from 3.90 to 1.40 and active review debt from 4.40 to 1.40, at 1.06x token cost and 1.09x runtime. Mechanism ablations show that removing either the hostile critic or owner-aware repair routing eliminates these gains. The resulting claim is deliberately narrow but consequential: under fixed artifact conditions, submission-path control materially improves reviewer-facing paper readiness even without claiming broader autonomous-discovery superiority.

1 Introduction

Autonomous research systems are improving rapidly. Recent open-source projects have shown that large language models can propose research ideas, write code, run experiments, and assemble paper drafts with much less human intervention than before. However, moving from a runnable pipeline to a submission-aware paper package remains difficult. Current systems can often produce a plausible manuscript before they can justify a defensible one. That gap is not merely about better models or better prose. It is a control problem: the system needs a disciplined way to decide what can be promoted, what must stay blocked, and how reviewer-facing defects are routed back into repair.

This paper studies a narrower late-stage systems question than end-to-end autonomous discovery: once experiments, claims, and draft text already exist, how should a paper candidate move through a reviewer-facing control loop? Our answer is an artifact-routed control architecture in which promotion, critique, and repair operate over explicit artifacts and owners rather than as a monolithic rewrite loop. Weak evidence routes to experiment owners, oversold claims route to storyline owners, figure debt routes to figure owners, and skeptical criticism remains read-only pressure rather than hidden drafting assistance. In other words, the central object is not an all-purpose writing agent, but a submission controller that decides which artifacts may advance, which must be blocked, and which owner must absorb each repair. The practical claim is modest but important: even if discovery

quality is unchanged, better submission-path control can still materially change whether a draft is reviewer-defensible.

The intended novelty is not simply “more reviewers” or “more agents.” The contribution is a control policy that couples three ingredients: artifact-gated manuscript promotion, read-only adversarial review, and owner-routed repair. Together, these ingredients specify what is allowed to advance, what must be blocked, and who is responsible for clearing each blocker. The closest counterfactual is not “the same system with fewer personas,” but a pipeline in which critique can directly rewrite the draft or unsupported claims can still be promoted without a visible owner transition. In that sense, XScientist Board is closer to a submission-path operating policy than to an agent ensemble. We instantiate this view in XScientist Board, which extends AI Scientist-style automation into a reviewer-facing submission-hardening loop with explicit success criteria, claim-evidence bindings, and structured hostile-critic findings.

Our scope is intentionally late-stage and control-oriented. We assume a pipeline can already produce candidate experiments, draft text, and review artifacts; the systems question here is how those artifacts should be admitted, challenged, and repaired on the path to submission. This is different from asking whether the underlying model already achieves state-of-the-art autonomous discovery.

We therefore evaluate the current snapshot under a deliberately bounded evidence boundary. The main empirical layer is a frozen artifact replay benchmark that measures submission-facing outcomes on the same topic and the same frozen idea realization across workflows and ablations. The secondary layer consists of executable implementation checks, a minimal routed-critique case, and synthetic routing-coverage tests. The release package also includes a stricter live shared-topic protocol plus a small three-topic local live pilot in the supplement, but the main paper’s empirical claims remain anchored to replay rather than to that pilot. The result is still a bounded paper, but it now carries auditable paired evidence: it argues for a submission-control architecture whose tested-suite replay behavior can already be audited under reviewer-visible defects.

2 Contributions and Scope

We study a specific systems question: what control structure can make autonomous research pipelines more defensible under reviewer scrutiny? We focus on the system architecture, implementation snapshot, and artifact-level control loop needed for more defensible autonomous paper packaging. We do not claim that the present version already establishes state-of-the-art autonomous discovery performance or benchmark-level paper-quality gains. This paper makes three concrete contributions:

1. We formulate submission hardening as an artifact-routed control-policy problem, in which manuscript promotion and repair are governed by explicit artifact states, blockers, and ownership transitions.
2. We specify the structured contracts required to make that control loop executable, including research plans, claim-evidence bindings, critic findings, and repair ownership.
3. We report a bounded empirical validation snapshot, including a populated frozen 10-topic artifact replay benchmark with workflow and ablation rows, a 46-test local validation suite, a reproducible minimal hostile-critic routing case, and a synthetic hostile-critic routing-coverage benchmark.

To avoid overclaiming, the draft uses an explicit claim ladder. Workflow extraction and targeted tests support implementation facts about what the system exposes. The frozen artifact replay suite

System family	Gate	RO critic	Owner repair	Block	Frozen eval
AI Scientist-style pipelines	No	No	Partial	No	No
Writing stacks	No	No	No	No	No
LLM review assistants	No	Partial	No	No	No
XScientist Board	Yes	Yes	Yes	Yes	Yes

Table 1: High-level positioning by control-policy feature. The distinction is not persona count, but which authority boundaries are explicit and auditable.

supports bounded tested-suite claims about submission hardening under fixed topics, fixed ideas, and fixed artifact packets. The minimal routed-critique case and routing-coverage stress test support behavioral claims about artifact routing under reviewer-visible defects. The three-topic local live pilot reported only in the supplement supports executability and packet-compilation claims for the live protocol, not stronger outcome claims. Stronger outcome claims about live end-to-end paper quality and external readiness calibration therefore remain withheld until a larger provider-backed shared-topic benchmark exists.

3 Related Work

Our work sits at the intersection of tool-using language agents, autonomous experimentation, critique-driven self-improvement, and AI-assisted scientific writing. ReAct and Toolformer show the value of interleaving reasoning with external actions and tool calls rather than relying on pure free-form generation [11, 8]. The AI Scientist is the closest academic antecedent for end-to-end autonomous research from ideation to paper drafting [5]. AI Scientist v2 moves closer to workshop-level autonomous submission through agentic branching and stronger review-stage automation [10], while recent benchmarking efforts such as MLAGentBench and MLGym highlight how difficult it is to evaluate language agents on realistic ML experimentation loops [1, 6]. Reflexion shows that explicit verbal feedback can improve agent behavior through self-critique [9]. ReviewerGPT suggests that LLMs can assist scientific review, but also underscores the need for grounded artifacts and reviewer-visible traceability [4]. On the implementation side, autoresearch, awesome-ai-research-writing, and DeepReviewer-v2 shaped our design space around research-program discipline, fixed-budget experimental logging, writing-skill decomposition, and structured review [2, 3, 7]. We build on these lines of work, but our emphasis is narrower: not how to generate more research activity, but how to govern what a system is allowed to promote as a submission.

Our position is that these lines of work are complementary, but they do not solve the same control problem. Relative to AI Scientist-style systems, our target is not broader ideation throughput or open-ended discovery, but submission-path control under reviewer-facing pressure. Relative to self-critique and reviewer-assistance systems, our hostile critic is read-only and cannot revise the manuscript directly. Relative to writing stacks, our central mechanism is not better phrasing alone, but artifact-gated promotion plus owner-routed repair. The novelty claim should therefore be read as a governance claim over artifact transitions, not as a claim that simply adding more roles is sufficient. This distinction is important for evaluation as well: a system can be strong at ideation or drafting yet still be weak at deciding what should count as submission-ready evidence.

Table 1 should be read as a positioning aid rather than a leaderboard. The point is not that adjacent systems can never instantiate pieces of this behavior, but that XScientist Board makes explicit promotion gating, read-only criticism, owner-routed repair, submission blocking, and frozen matched-idea evaluation part of the primary control object.

Artifacts (state):	plan, claim–evidence bindings, run logs, tables, manuscript state.
Promotion gate (policy):	only promote claims whose artifact bindings and blocker status satisfy the current promotion rules.
Review (pressure):	reviewer board + <i>read-only</i> hostile critic emit structured issues (no direct manuscript edits).
Routing (ownership):	each issue → an owner lane (experiments / figures / storyline / LaTeX), with verification steps.
Repair (transition):	owners update artifacts; the gate recomputes readiness and promotion eligibility.

Figure 1: XScientist Board treats paper finalization as a control policy over artifact state transitions. The critic lane is intentionally read-only: it can block promotion and create repair tasks, but it cannot rewrite the manuscript.

4 Problem Setting and Design Goals

We consider the problem of submission-aware autonomous research production. The input is a research topic or seed idea. The desired output is not merely a draft manuscript, but a package of artifacts that can support reviewer-facing submission packaging: a research program with explicit budgets, experiment records that preserve both successful and failed runs, figures and tables linked to concrete claims, a manuscript whose claims are bounded by evidence, review and critic artifacts that expose unresolved debt, and a final readiness judgment that can block low-integrity submission. We assume that candidate text, experiment traces, and review artifacts may already exist, but may be weakly coupled, partially unsupported, or difficult to audit. The core control problem is therefore deciding what can enter the promoted narrative and what must remain blocked as debt. Unlike pure automated discovery, the objective is not simply to maximize novelty or branch diversity, but to jointly optimize evidence quality, review robustness, repairability, and operational transparency. The paper does not claim to verify scientific truth; it claims to control what may be promoted under the currently available artifact state.

These requirements translate into four design goals:

1. Do not hide process debt.
2. Do not let writing compensate for missing evidence.
3. Do not let critique mutate the paper directly.
4. Do not conflate exploration with submission.

5 System Overview

XScientist Board organizes the pipeline as a multi-agent board with explicit lanes: planner, experiment manager, experiment workers, results analyst, storyline editor, LaTeX hygiene editor, humanizer, quality gate, reviewer board, hostile critic, and repair lane. Figure 1 summarizes the resulting control loop.

The board structure matters because different agents should optimize for different objectives and hold different authority. The experiment manager should maximize evidence value under budget. The writing lanes should maximize clarity without inventing support. The hostile critic should maximize the probability of catching weak claims before an external reviewer does, but without editing the paper directly. At design level, the architecture is constrained by four invariants: unsupported claims should not advance without artifact bindings; writing should not erase experimental debt;

Failure mode	System mechanism	Intended block or routing behavior
Claim inflation	Claim-evidence graph + hostile critic	Stop unsupported claims from entering the promoted narrative without a bound experiment and artifact path
Weak or missing evidence	Research plan + quality gate + repair routing	Send evidence debt back to experiment owners instead of rewriting around it
Figure or caption drift	Figure specification + manuscript state	Block promotion when lead visuals are not traceable to the stated claim path
Silent process debt	Strict fallback discipline + stage standards	Surface hidden heuristics, blocked stages, and unresolved debt before submission packaging

Table 2: Failure modes that motivate the artifact-routed control structure.

critique should remain read-only; and every blocking issue should expose a concrete owner and verification path.

Authority boundaries matter at least as much as lane count. Experiment lanes may create or verify evidence but cannot promote unsupported claims. Writing lanes may rewrite narrative but cannot change the evidence state. Reviewer lanes may open or clear issues but cannot silently mutate the manuscript. The hostile critic may block promotion but is read-only. The quality gate may accept or reject promotion but does not author new claims. These boundaries are what turn the board from a collection of personas into a control policy.

6 Artifact Contracts

The central design principle in XScientist Board is that every major stage produces a structured artifact.

The planning layer produces a research program and machine-readable plan with owners, dependencies, success criteria, kill criteria, and branch outcomes. The claim-evidence graph links hypotheses, experiment tasks, metrics, claims, figures, and limitations, constraining both writing and repair. The figure specification and manuscript state track whether visuals and section-level claims are actually ready for promotion. The review layer persists a running issue ledger, hostile-critic findings, ownership-aware repair tasks, and cross-run self-evolution summaries. Together these artifacts define a promotion policy over paper state, not just a logging scheme around a drafting loop.

In the intended submission configuration, a candidate should not be promoted to the final manuscript package unless the following conditions hold:

1. each lead claim is mapped to a baseline-comparable experiment record;
2. each lead claim is covered by at least one figure or table path;
3. no unresolved major blocker remains without an owner and verification path;
4. the hostile critic has no active blocking issue that remains unaddressed;

5. strict fallback debt is either zero or explicitly disclosed and accepted.

These criteria matter because the dominant failure mode in autonomous paper systems is often not inactivity, but premature promotion of under-supported artifacts into the submission path. The architecture is only novel if these routing constraints genuinely change what is allowed to ship.

7 Hostile Critic Protocol

Most self-improving paper pipelines treat critique as a softer version of revision. We instead model a read-only adversarial review lane whose explicit goal is to identify the strongest supported rejection argument. The implementation instantiates that lane with specialized review roles covering claim-evidence consistency, reproducibility, statistical adequacy, related-work positioning, meta-review synthesis, and desk-reject screening. Each finding must include an issue identifier, severity, blocker class, target type and identifier, an attack angle, a short failure explanation, evidence references, and suggested verification steps. This keeps the critic from drifting into vague negativity and keeps repair grounded.

8 Review and Repair as a Routing Problem

After normal review and hostile criticism, XScientist Board converts active issues into repair tasks. Repair is not treated as a generic rewrite step. Evidence holes and reproducibility gaps go to experiment owners, figure and caption issues go to figure owners, oversell and novelty-framing issues go to the storyline editor, clarity problems go to writing owners, and unresolved routing problems return to the planner. This design is intended to prevent a common pathology in autonomous writing systems: asking a writing model to solve fundamentally experimental weaknesses through wording changes.

9 Empirical Evaluation Under a Frozen Artifact Replay Boundary

We implemented the proposed architecture in the XScientist codebase as a concrete workflow mode rather than a purely conceptual design. This section therefore reports a bounded empirical study rather than a live end-to-end discovery benchmark. The main empirical layer is a frozen artifact replay suite that keeps the topic, one idea realization, and the reviewer-facing artifact schema fixed across workflows and ablations. The secondary layer is a traced routed-critique case plus synthetic routing-coverage checks. This combination is designed to test the contribution the paper actually claims most strongly: submission hardening over already-instantiated artifact packets.

Evaluation scope. The evaluation in this draft addresses four narrower questions: under fixed topics, fixed ideas, and fixed artifact packets, do richer control policies improve submission-facing metrics; do hostile criticism and owner-aware repair routing matter as mechanisms on that same frozen replay suite; do hostile-critic findings become structured owner-routed repair work on a real artifact packet; and does routed critique behavior persist across more than one injected failure pattern? A three-topic live pilot is included only in the supplement as a smoke test that the append-only live protocol, ledger compilation, and table rendering execute end-to-end on this machine. The main paper still does not claim live end-to-end paper-quality gains, critic utility on launched runs, or external calibration of the readiness signal.

Workflow	Topics	Paired	Verdict	Decision	Ready Rate	Unsupported Claims	Active Review Debt	Critic Blockers	Cost Ratio	Runtime Ratio
classic_pipeline	10	0/0	baseline	withhold	0.00	3.90	4.40	1.30	1.00x	1.00x
agentic_tree	10	10/10	tradeoff	tradeoff	0.00	3.40	4.10	1.30	1.20x	1.14x
program_driven	10	10/10	worth_it	keep	0.50	2.20	2.50	0.90	1.03x	1.06x
review_board	10	10/10	tradeoff	tradeoff	0.60	2.40	2.40	0.30	1.12x	1.12x
multi_agent_board	10	10/10	worth_it	keep	0.80	1.40	1.40	0.20	1.06x	1.09x

Table 3: Frozen artifact replay workflow comparison under a 10-topic matched-idea manifest and matched budget policy.

Frozen artifact replay workflow comparison. The replay suite instantiates ten frozen topics, freezes one idea realization per topic, and materializes workflow-specific artifact packets under the same metric schema. Every workflow and ablation row is compiled from the same manifest path, and the resulting consistency audit is fully clean (90/90 run references with zero fixture mismatches). This is not a substitute for a live provider-backed benchmark, but it is real paired evidence for the submission-hardening layer. Table 3 summarizes the workflow comparison.

Metrics. Each topic produces a reviewer-facing artifact packet (quality summary, review ledger, critic findings, and repair plan artifacts) from which we extract submission-hardening metrics. *Ready rate* is the fraction of topics whose artifact-derived `submission_readiness.ready` flag is true. *Unsupported claims* is the mean count of promoted manuscript claims without required evidence bindings (`unsupported_claims_count`). *Active review debt* is the mean count of unresolved issues in the persisted review ledger. *Critic blockers* is the mean count of currently blocking hostile-critic findings (`blocking_issue_count`). Token cost and runtime are reported as ratios against `classic_pipeline` using manifest-recorded token totals and wall-clock minutes. The full metric schema and extraction path are documented in the supplement.

Workflows. We compare five workflow modes that differ in their control policy rather than in topic/idea content. `classic_pipeline` is the baseline draft-and-review loop without promotion gating. `agentic_tree` adds broader agentic search but no submission-path governance. `program_driven` adds explicit program discipline and a repair lane. `review_board` adds multi-role review and repair. `multi_agent_board` instantiates the full submission-hardening loop with artifact-gated promotion, a read-only hostile critic, and owner-aware repair routing. Exact lane composition is extracted from the runnable implementation and recorded in the supplement.

Figure 2 visualizes the per-topic deltas behind the strongest row, showing that most topics move in the jointly improved (negative, negative) direction for unsupported claims and review debt. Three reviewer-relevant patterns then emerge. First, extra search alone is not enough: `agentic_tree` records zero ready wins, only partial unsupported-claim relief (5/10 wins and 5/10 ties), and a 1.20x/1.14x token-runtime premium, so it fails the simplicity-aware bar. Second, not every richer workflow becomes a clean win: `review_board` delivers strong paired gains (6/10 ready wins, 9/10 unsupported-claim wins, and 10/10 review-debt wins), but its 1.12x token and runtime ratios still keep it in the `tradeoff` bucket. Third, the positive empirical story is selective rather than indiscriminate: `program_driven` and `multi_agent_board` are the only `worth_it` rows, and `multi_agent_board` is the strongest row overall. This selectivity matters rhetorically: the paper is not arguing that more orchestration is always better, but that a particular submission-control policy can clear reviewer-facing metrics under fixed conditions.

Relative to `classic_pipeline`, `multi_agent_board` improves ready rate on 8/10 topics with 2 ties and no losses ($p = 0.004$), and improves unsupported claims and active review debt on all 10/10 topics (both $p < 0.001$). The mean shifts remain large: ready rises from 0.00 to 0.80, unsupported

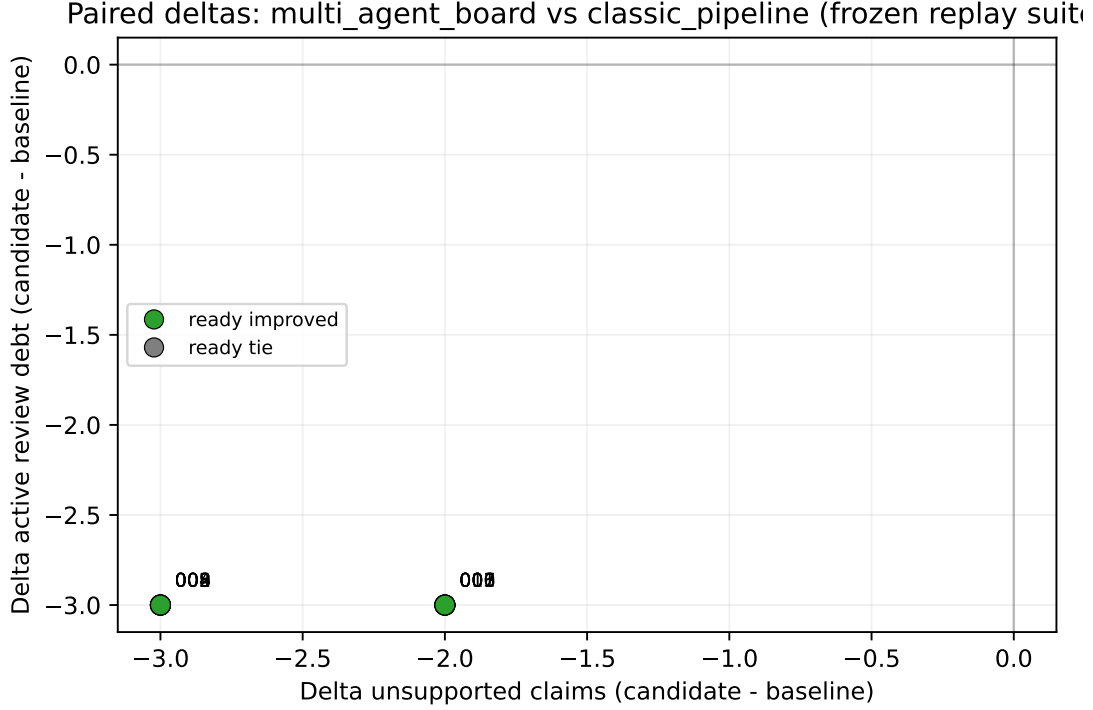
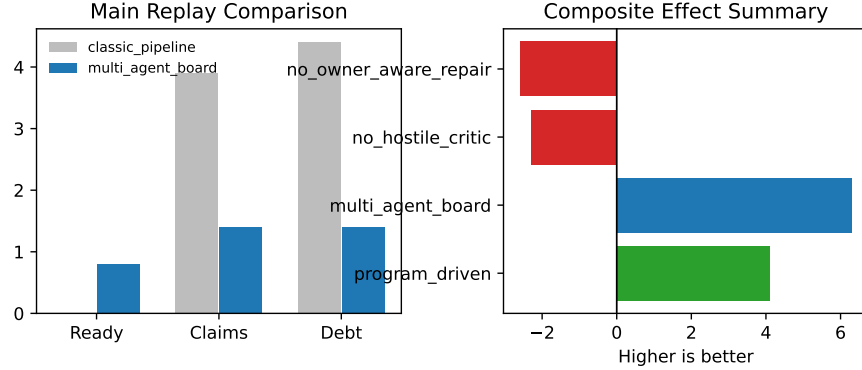


Figure 2: Per-topic paired deltas for `multi_agent_board` versus `classic_pipeline` on the frozen artifact replay suite. Each point is a topic; the axes show changes in unsupported-claim count and active review debt (candidate minus baseline; lower is better). Color indicates whether readiness improved (green) or tied (gray).

claims fall from 3.90 to 1.40, and active review debt falls from 4.40 to 1.40, while token and runtime ratios remain 1.06x and 1.09x. Supplementary leave-one-topic-out, bootstrap, burden-split, and cost-benefit checks preserve the same favorable replay direction, reducing the risk that the result is carried by one topic or one metric weighting. More importantly, the workflow remains strongest under stricter topic-level criteria: direct pairwise-dominance and raw residual-dominance analyses both keep `multi_agent_board` ahead on all 10/10 topics, with a mean strict-bundle winner margin of 2.90 failed thresholds. Archetype analyses show why: `review_board` is usually the nearest competitor on moderate and transition topics, but the full board still closes 1.00 of baseline strict-bundle burden on the moderate cluster and 0.86 even on the severe blocker-heavy cluster, where the strongest non-board closure is only 0.75. Transition-side analyses localize that edge to the final `review_board`→`multi_agent_board` step: on moderate topics it removes the canonical {claims, debt, repair} near-pass packet on all 6/6 topics, and on blocker-heavy topics it supplies the decisive blocked-stage and repair lift that earlier steps do not. The hard tail is therefore not just smaller but structurally different under the full board: `review_board` leaves the same six-threshold packet on all three blocker-heavy topics, whereas the full board breaks that tail into three smaller signatures, shrinks the shared severe core from six threshold families to one, and keeps a mean pairwise packet overlap of only 0.45. These rows still support only a bounded replay claim, but they show that on this frozen suite the full board clears reviewer-critical metrics topic by topic without a large extra-cost penalty.

Submission-Hardening Evidence Summary



Replay evidence summary

- multi_agent_board vs classic_pipeline: ready 0.00 -> 0.80, claims 3.90 -> 1.40, debt 4.00 -> 1.00
- paired support verdict: worth it / decision: keep
- program_driven remains a positive lower-cost row: token ratio 1.03x, runtime ratio 1.06x
- hostile-critic ablation stays harmful: delta ready -0.30, delta claims 0.60, delta debt 0.00
- repair-routing ablation stays harmful: delta ready -0.80, delta repair-ready -0.291, delta repair-ready -0.291, delta repair-ready -0.291
- supplementary packet also includes robustness, stratification, frontier, hardest-topic,

Figure 3: Compact submission-hardening evidence summary. The panel compresses the strongest workflow comparison, the promoted-versus-harmful composite effects, and the key replay takeaways into a single reviewer-facing visual overview.

Mechanism ablations. The replay suite also allows the paper to test whether the most novel mechanisms are actually doing work rather than merely increasing agent count. Tables 4 and 5 summarize the results.

Removing the hostile critic produces topic-wise regressions rather than a mere average-case dip: ready never wins (0/7/3, $p = 0.125$), unsupported claims worsen on 5/10 topics ($p = 0.031$), and active review debt worsens on all 10/10 topics ($p < 0.001$). Removing owner-aware repair routing is harsher still: ready loses on 8/10 topics ($p = 0.004$), active review debt worsens on 6/10 topics ($p = 0.016$), repair readiness drops by 0.29, and blocked stages rise by 1.50. Supplementary robustness checks keep both ablation directions harmful, and topic-level ablation residual-dominance keeps the intact board as the unique lowest-failure setting on all 10/10 topics, with a mean margin of 1.80 failed thresholds over the nearest ablation. Archetype analyses also show a clean mechanism split: hostile-critic removal mainly reopens debt and claims control on moderate topics, whereas owner-aware routing is what most strongly breaks readiness, blocked stages, and repair readiness; operationally, the intact board turns the moderate cluster into 6/6 exact strict-bundle passes, but the two ablations fall to 0/6 and preserve only 5/6 and 2/6 within-two near-passes. On the severe blocker-heavy cluster, the same split becomes sharper. Relative to **review.board**, the intact board cuts mean residual burden from 6.00 failed thresholds per topic to 3.33, but the ablations rebound

Setting	Topics	Paired	Verdict	Decision	Ready Rate	Unsupported Claims	Active Review Debt	Critic Blockers	Cost Ratio	Runtime Ratio
full_board	10	0/0	baseline	withhold	0.80	1.40	1.40	0.20	1.00x	1.00x
no_hostile_critic	10	10/10	no_gain	discard	0.50	2.00	2.80	0.50	0.95x	0.96x

Table 4: Hostile-critic ablation on the frozen artifact replay suite under matched idea fixtures and matched budget policy.

Setting	Topics	Paired	Verdict	Decision	Ready Rate	Repair Verification Readiness	Active Review Debt	Blocked Stages	Cost Ratio	Runtime Ratio
full_board	10	0/0	baseline	withhold	0.80	0.96	1.40	0.60	1.00x	1.00x
no_owner_aware_repair	10	10/10	no_gain	discard	0.00	0.67	2.00	2.10	0.97x	0.97x

Table 5: Owner-aware repair-routing ablation on the frozen artifact replay suite under matched idea fixtures and matched budget policy.

to 5.00 and 5.67. That rollback is not diffuse: both ablations have severe rollback precision 1.00, so every reopened threshold belongs to the exact set the final board step had removed, while recall still rises to 0.625 and 0.875. The raw metrics tell the same story: the full board raises repair readiness from 0.77 to 0.93 while cutting debt from 3.00 to 2.00 and blocked stages from 2.67 to 1.33, but hostile-critic removal gives back the debt and repair lift and routing removal gives back the blocked-stage and repair lift. Normalizing against the intact board’s own severe-topic gains sharpens the contrast further: hostile-critic removal preserves only 0.18 of average hard-tail gain and owner-aware routing falls to -0.04; at the topic-metric level, the two ablations preserve only 7/13 and 3/13 positive severe gains, and across all 18 severe topic-metric cells neither one is uniquely best even once, whereas the intact board is uniquely best on 7/18. These remain bounded replay-suite mechanism claims, but they make the paper’s ablation story substantially stronger than a generic “the critic helps” narrative.

Minimal hostile critic routing case. We constructed a minimal project state containing a lead claim, a figure, and a manuscript shell, then executed the hostile critic through the persisted review-job interface. The resulting artifact packet is summarized in Table 6. This micro-case remains valuable because it exposes the concrete routing behavior behind the replay numbers: reviewer-visible defects become persisted debt with named owners, produced artifacts, and verification-ready repair work rather than free-form rewrite advice.

Synthetic hostile-critic routing coverage. To move beyond a single anecdotal example, we also ran a four-scenario synthetic routing-coverage benchmark covering **oversell_only**, **repro_gap**, **figure_gap**, and **mixed_attack**. Scenario-level outputs are shown in Table 7. Across these cases, the hostile critic produced an average of 2.75 findings and 1.75 blocking findings per scenario. All 4/4 scenarios ended with submission readiness blocked, which should be interpreted as conservative sensitivity under injected failure patterns rather than as utility on final paper outcomes. The induced repair plans routed to three distinct owners overall: **experiment_agent**, **figure_agent**, and **storyline_editor**.

Workflow distinctiveness and implementation robustness. Supporting implementation evidence comes from two additional checks. First, executable workflow extraction shows that only **multi_agent_board** currently combines an independent critic lane, non-zero critic roles, and strict fallback discipline; the full capability table is deferred to the supplement. Second, we re-executed a targeted validation suite of 46 unit tests in the local environment (`python -m unittest ...`). The passing suite spans workflow defaults, runtime plans, planning contracts, critic persistence, repair

Active	Blocking	Top blocker	Repair tasks	Routed owners	Verify-ready
3	2	oversell	3	figure_agent, storyline_editor	1.0

Table 6: Minimal routed-critique micro-case from the current implementation snapshot.

Scenario	Findings	Blocking	Submission ready
oversell_only	2	1	No
repro_gap	3	2	No
figure_gap	2	1	No
mixed_attack	4	3	No

Table 7: Synthetic hostile-critic routing-coverage benchmark. The blocked outcomes should be read as conservative sensitivity under injected failure patterns, not as utility on final paper quality.

planning, source planning, stage standards, process alignment, readiness benchmarking, run-index reporting, and research-pipeline artifact persistence.

What this evidence does and does not show. Taken together, these evaluations support seven bounded statements: the architecture is implemented; routed critique persists as structured debt; repair ownership is routable; workflow differentiation is executable; artifact-derived readiness can be computed reproducibly; the frozen replay suite shows tested-suite gains for `program_driven` and `multi_agent_board`; and replay ablations show that hostile criticism and owner-aware repair routing are empirically useful on the tested suite. The supplementary robustness, stratification, case-study, and metric-correlation checks make that replay packet harder to dismiss as either a single-topic artifact or a one-metric story. They still do not show large-scale live paper-quality gains on launched autonomous papers, critic utility on provider-backed runs, or predictive validity of the readiness heuristic. The present paper therefore supports a bounded tested-suite submission-hardening claim, not benchmark-outcome superiority in the broadest sense.

10 Live Benchmark Discipline Beyond the Current Snapshot

Late-stage autonomous paper systems are unusually easy to over-evaluate after the fact: topics can drift, ideation realizations can change across workflows, crashes can disappear between runs and tables, and pooled means can hide same-topic regressions. To reduce those degrees of freedom, the release package still includes a stricter live shared-topic protocol adapted from autoresearch’s fixed-budget, single-metric, append-only evaluation discipline. That protocol freezes the topic suite before outcomes are inspected, freezes one shared idea realization per topic, runs the baseline workflow first, preserves an append-only execution ledger with explicit crash evidence, compiles `keep/discard/crash` verdicts for realized runs, and evaluates richer workflows with paired deltas plus simplicity-aware cost tradeoffs rather than pooled means alone.

The new replay suite should therefore be read as a local empirical bridge, not a replacement for the live protocol. It gives the current draft real paired rows and mechanism ablations under frozen conditions, while the live shared-topic protocol remains the next step for stronger end-to-end claims. The supplementary material now also reports a three-topic local live pilot that exercises this protocol end-to-end, but that pilot is intentionally treated as an executability smoke test rather than as the basis for the paper’s main outcome claims. This split keeps the paper empirical without pretending that the decisive live shared-topic benchmark has already been populated.

11 Limitations

The framework increases runtime and token cost. A hostile critic can become an infinite rejection engine if stop conditions or escalation rules are poorly designed. The quality of routing depends on the quality of claim-evidence bindings; weak bindings can produce noisy repair plans and false blockers. Internal artifact metrics can also drift away from external paper quality if they are optimized too aggressively, especially if future systems learn to game readiness-facing heuristics without improving the underlying paper. Most importantly, the current benchmark is still dominated by a frozen artifact replay suite rather than a decisive live provider-backed shared-topic launch. The three-topic local live pilot in the supplement reduces uncertainty about executability, but it is too small and too local-model-specific to resolve the external-validity question. The present evidence can therefore defend bounded tested-suite claims about the submission-hardening layer, but it cannot yet defend live end-to-end paper-quality gains, external readiness calibration, or acceptance prediction. The longer-term contribution should remain tied to whatever the later live benchmark actually supports.

12 Conclusion

We presented XScientist Board, an artifact-routed control architecture for autonomous research hardening. The key idea is to treat paper production as a board of specialized lanes connected by structured artifacts, explicit authority boundaries, and promotion rules. The main contribution is a submission controller over artifact state transitions: claims cannot advance without bindings, critics cannot rewrite, and repairs must name owners. By separating planning, evidence generation, writing, normal review, hostile criticism, and repair, the system is designed to surface silent failure debt and make autonomous paper pipelines easier to audit and repair.

The current evidence now supports a stronger but still bounded result than the earlier draft: on a frozen artifact replay suite, submission-path control improves reviewer-facing outcomes for `program.driven` and especially for `multi_agent.board`, while hostile criticism and owner-aware repair routing remain empirically useful mechanisms on that tested suite. The supplement further shows that the stricter live protocol can be exercised end-to-end in a small local pilot, which strengthens the artifact packet’s reproducibility story without changing the main claim boundary. The durable contribution of this draft is therefore both conceptual and operational: a control vocabulary for artifact-routed submission hardening plus concrete evidence that governing promotion, criticism, and repair as first-class artifact transitions can make autonomous papers more reviewer-defensible under fixed conditions.

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